

Micro I, class 8

We started a discussion of the theory of the firm. The firm chooses a *production plan*, $y \in \mathbb{R}^L$. Elements of a firm's plan can be indexed by date, location, or state of the world, of course, just as for the consumer. If $y_\ell > 0$, then the firm is producing good ℓ as a *net output*; if $y_\ell < 0$, then the firm is using good ℓ as a *net input*. The firm's profit at price $p \in \mathbb{R}_{++}^L$ is therefore $p \cdot y$. The constraint is that the plan y must lie in a set $Y \subset \mathbb{R}^L$ describing the firm's *technology*. The firm's problem is

$$(PM) \quad \pi(p) = \max_{y \in Y} p \cdot y.$$

We noted that this problem has the same mathematical structure as the expenditure minimization problem. Indeed (PM) is identical to $-\pi(p) = \min_{-y \in -Y} p \cdot (-y)$. The analogue of “raising $\bar{u}$ ” in the EM problem is replacing the production set Y with a strict subset of Y in PM (a worse technology). This already tells us that the value function for PM, π (the *profit function*), is *convex* and *homogenous of degree 0*.

We read off the properties of the profit functions and some implications of those for profit-maximizing net supplies. I outlined a generalization of the curvature results for value functions that we have been proving. It asserts in words that the maximum over a family of convex functions is convex.

We went on to discuss several assumptions often put on the technology: *closedness* (continuity); *no free lunch*; *possibility of inaction*; *free disposal* (monotonicity); *nonincreasing-*, *nondecreasing-*, and *constant-returns to scale*; and *convexity*. I left this as an open question: if $0 \in Y$ then what is the relationship between “ Y is convex,” and “ Y exhibits nonincreasing returns to scale?”

I then introduced the notion of a *single-output technology* and the associated *production function*; in the notation of the class, $f(z) = \max\{q \mid (q, -z) \in Y\}$. By a direct argument (or appeal to the Principle of Optimality), we can write the firm's profit maximization problem as $\max_{z \geq 0} pf(z) - w \cdot z$ (choose an input vector to maximize profit, with the output choice given by the production function).

Next up we will discuss various methods for comparative statics. I have notes on the topic, *Monotone Comparative Statics*, that you might want to consult. (It is not covered much in Kreps or MWG.)¹

¹As mentioned, I recommend chapter 10 of Sundaram's *A First Course in Optimization*, and Chapter 2 of Vives's book *Oligopoly Pricing* (available as an e-book from the ASU library). In addition you might look at succinct new paper by my former classmate, Rabah Amir, and David Rietzke “A Comment on: Monotone Comparative Statics,” *Econometrica* 93.4 (2025): 1481-1490, which sets out the basic question clearly; you might find their simple leading example useful. (Rabah, as with Xavier Vives, were pioneers in introducing economists to lattice methods.)